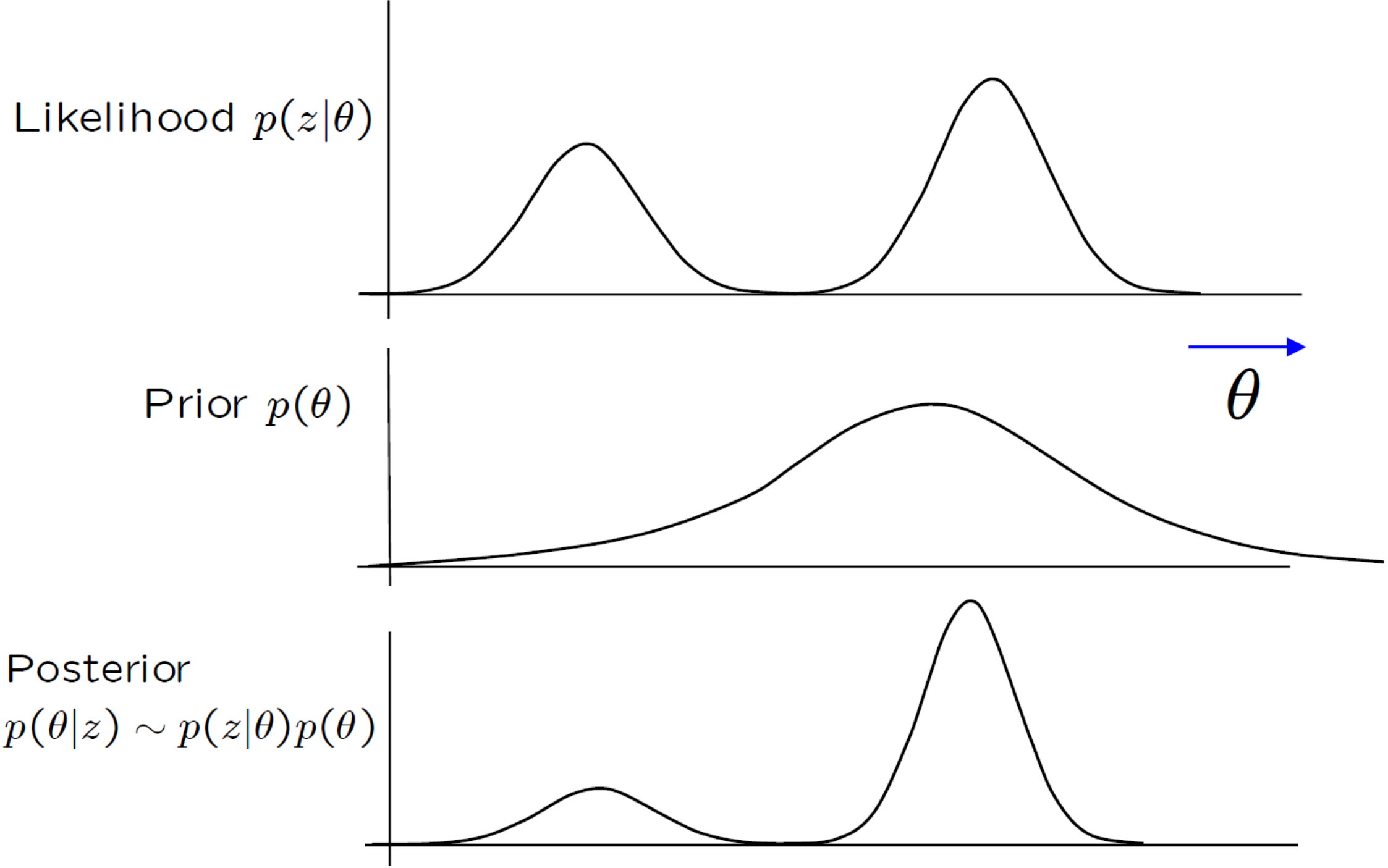
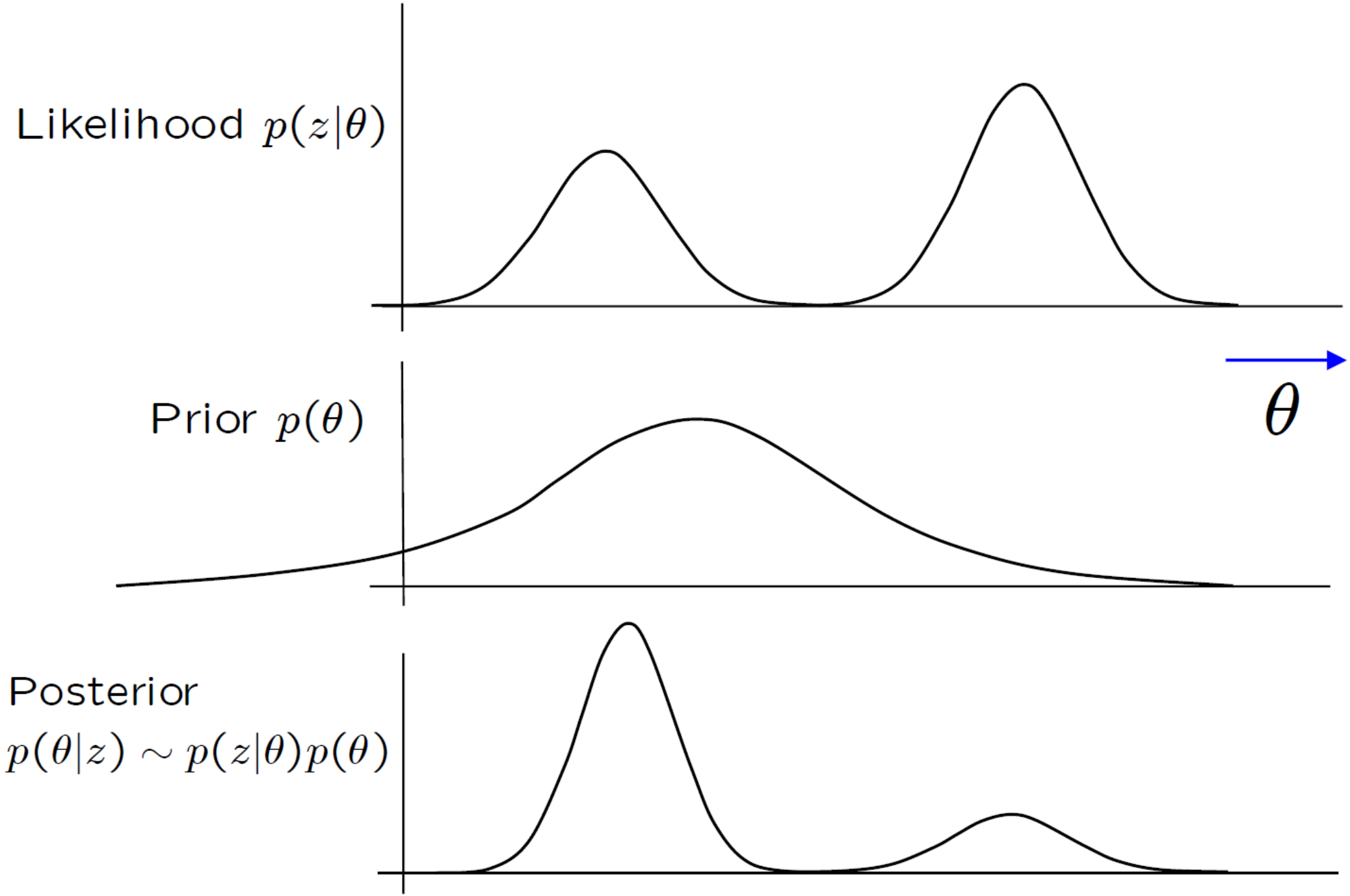
Maximum Likelihood Estimation vs Maximum A Posterior

Rejection, Importance sampling

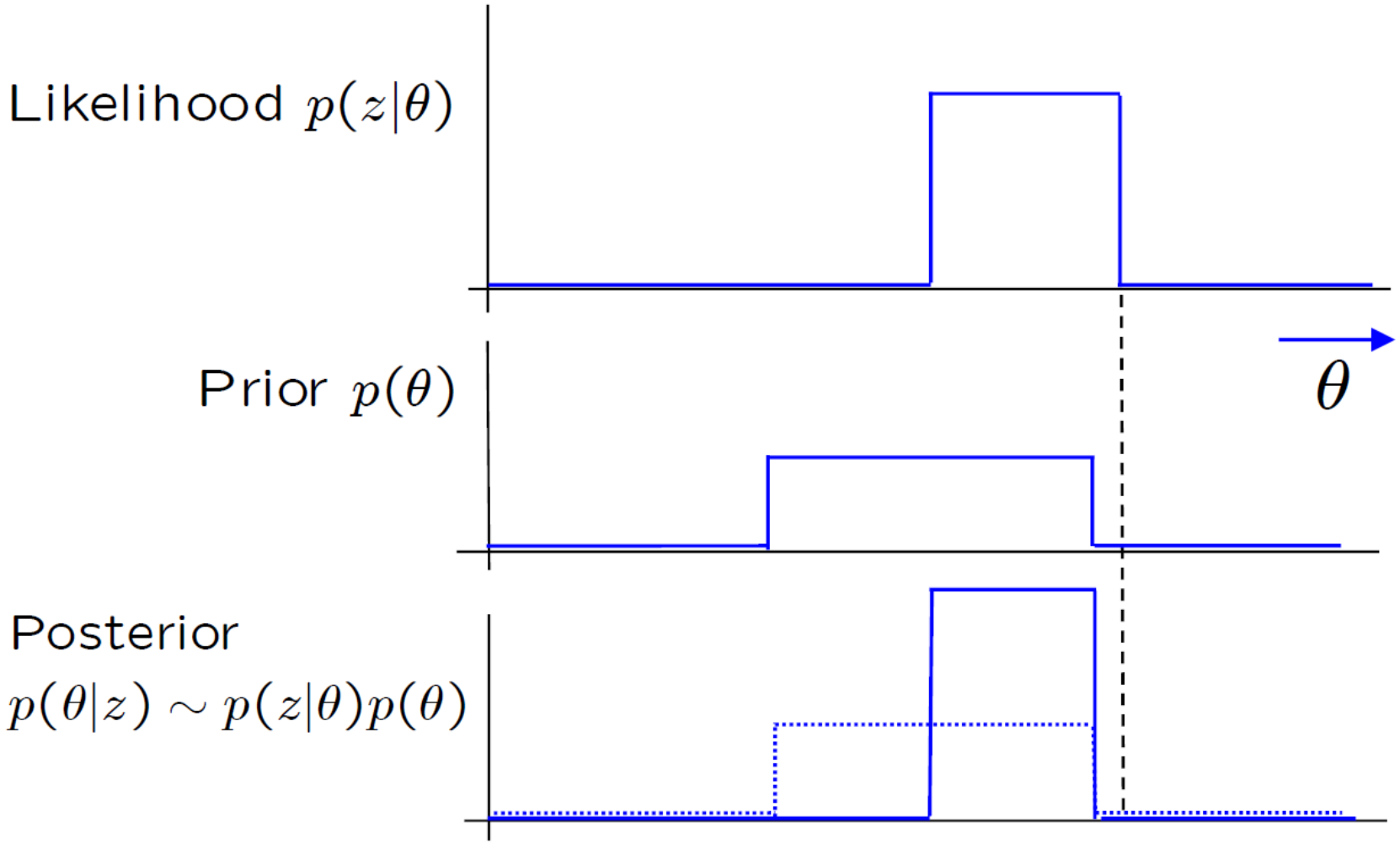
Use Bayes' rule to get the posterior density

$$p(\theta|z) = \frac{p(z|\theta)p(\theta)}{p(z)} = \frac{\text{likelihood} \times \text{prior}}{\text{normalizing factor}}$$


Likelihood $p(z|\theta)$
Prior $p(\theta)$
θ
Posterior
$p(\theta|z) \sim p(z|\theta)p(\theta)$


Likelihood $p(z|\theta)$
Prior $p(\theta)$
θ
Posterior
$p(\theta|z) \sim p(z|\theta)p(\theta)$

Example: posterior for uniform distributions



Maximum Likelihood Estimation

$$\theta_{MLE} = \arg\max_{\theta} \log P(X|\theta)$$

$$= \arg\max_{\theta} \log \prod_{i} P(x_i|\theta)$$

$$= \arg\max_{\theta} \sum_{i} \log P(x_i|\theta)$$

The only difference is that MAP includes prior in the formula, which means that the likelihood is weighted by the prior in MAP.

$$\theta_{MAP} = \arg\max_{\theta} P(X|\theta)P(\theta)$$

$$= \arg\max_{\theta} \log P(X|\theta) + \log P(\theta)$$

$$= \arg\max_{\theta} \log \prod_{i} P(x_i|\theta) + \log P(\theta)$$

$$= \arg\max_{\theta} \sum_{i} \log P(x_i|\theta) + \log P(\theta)$$

If we use the simplest prior in our MAP estimation, i.e. uniform prior. This means, we assign equal weights everywhere, on all possible values of the θ.

$$\theta_{MAP} = \arg\max_{\theta} \sum_i \log P(x_i|\theta) + \log P(\theta)$$

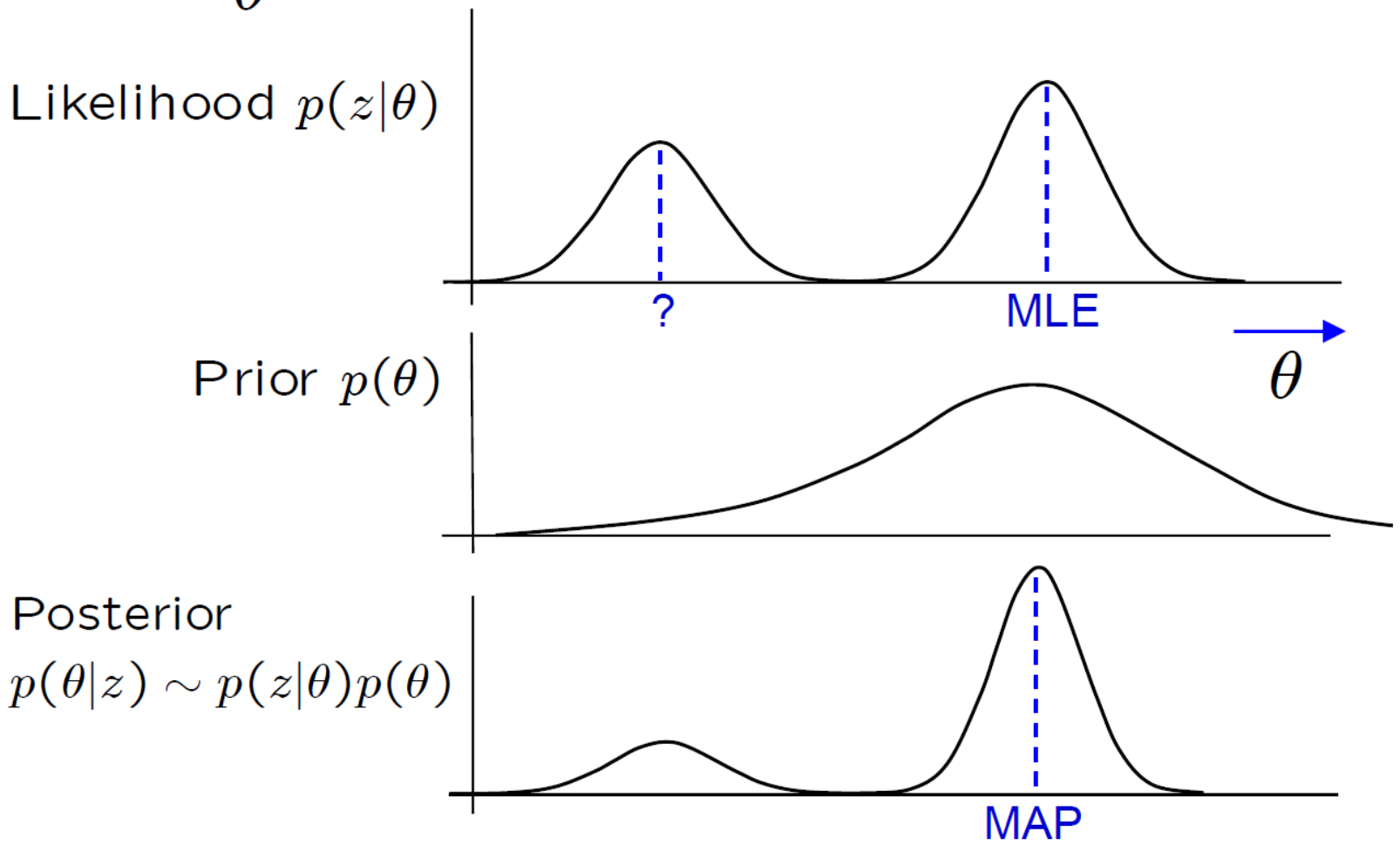
$$= \arg\max_{\theta} \sum_i \log P(x_i|\theta) + const$$

$$= \arg\max_{\theta} \sum_i \log P(x_i|\theta)$$

$$= \theta_{MLE}$$

MAP Estimator

$$\hat{\theta} = \arg\max_{\theta} p(\theta|\mathbf{z})$$



MLE: Posterior mode = Prior mode

- $P(\theta \mid X) \propto P(X \mid \theta)\, P(\theta)$
- $\log P(\theta \mid X) \propto \log P(X \mid \theta) + \log P(\theta)$
- Suppose θ^* is the posterior mode as well as prior mode
- $\log P(\theta^* \mid X) \propto \log P(X \mid \theta^*) + \log P(\theta^*)$
- Take the first derivative, and evaluate it at θ^*
- $\log P(\theta^* \mid X)' \propto \log P(X \mid \theta^*)' + \log P(\theta^*)'$
- Both $\log P(\theta^* \mid X)' = 0$ and $\log P(\theta^*)' = 0$ due to maximum (mode)
- So $\log P(X \mid \theta^*)' = 0$, and θ^* is the MLE
- PPM (prior posterior mode) Algorithm: iteratively update the prior mode (mean) using posterior mode (mean)

Buffon's Needle

Bayesian Analysis
Publisher: International Society for Bayesian Analysis

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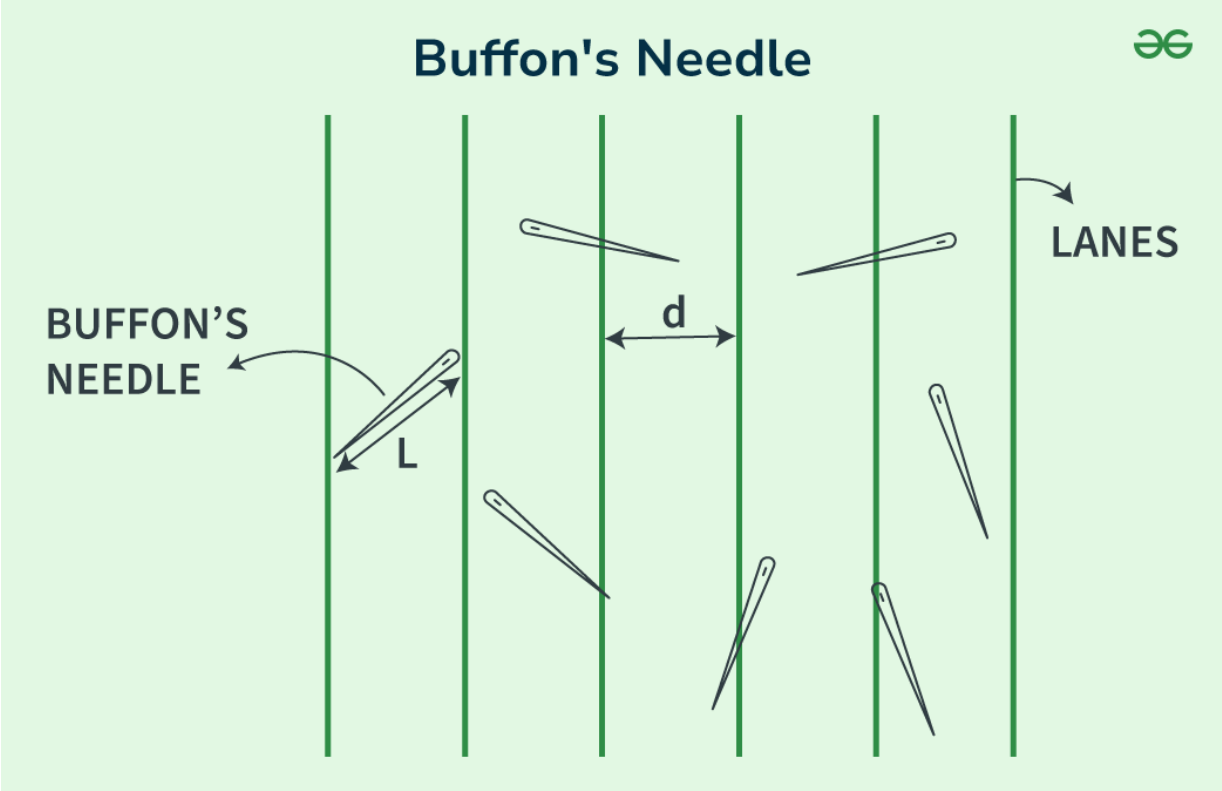
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Estimating π with Buffon's Needle

- Buffon's Needle is a probability problem first proposed by Georges-Louis Leclerc, Comte de Buffon, in 1733 and later published in 1777.



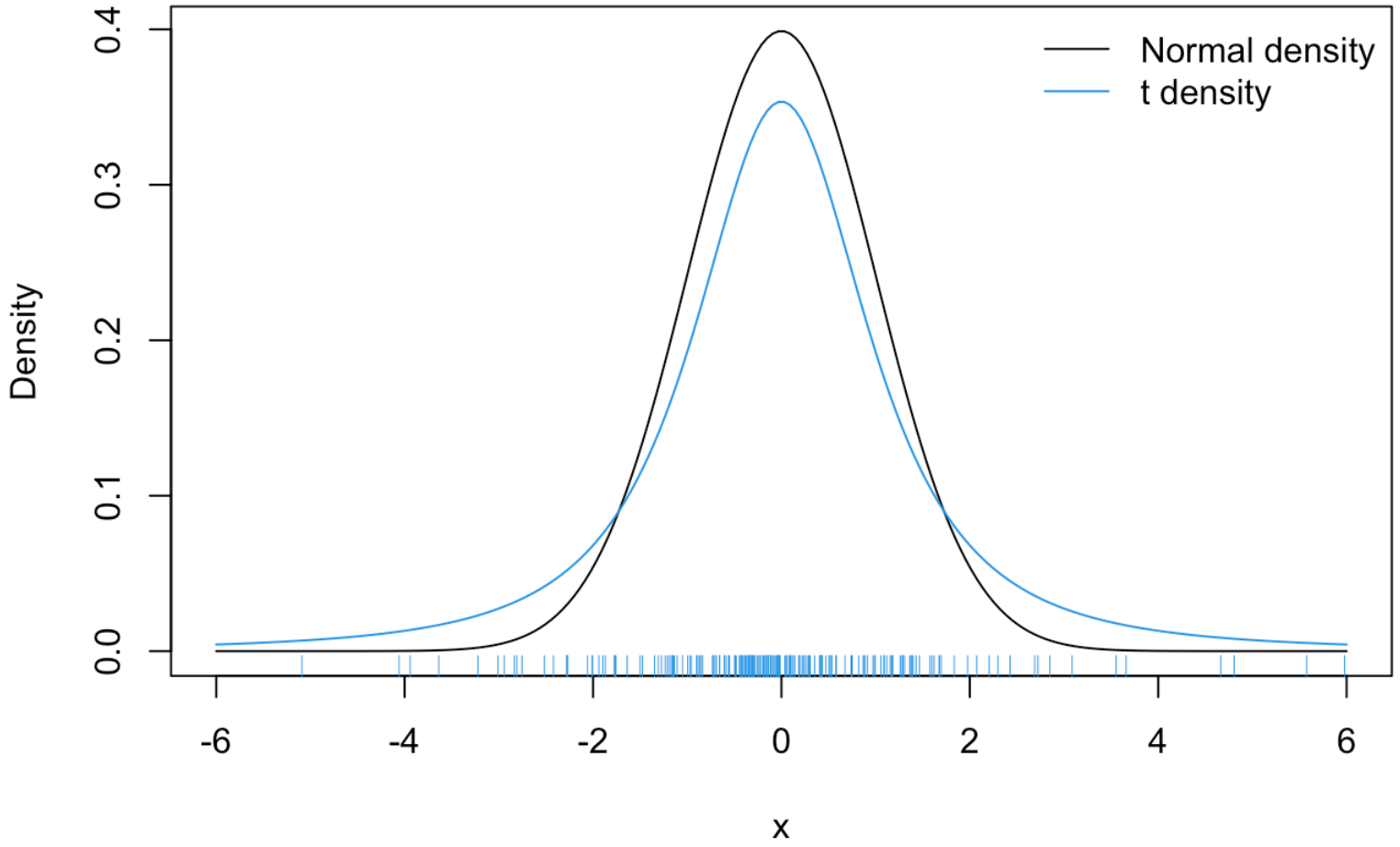
Estimating Pi

- The question is: **What is the probability that the needle will cross one of the lines?**
- *P = 2L/πd*

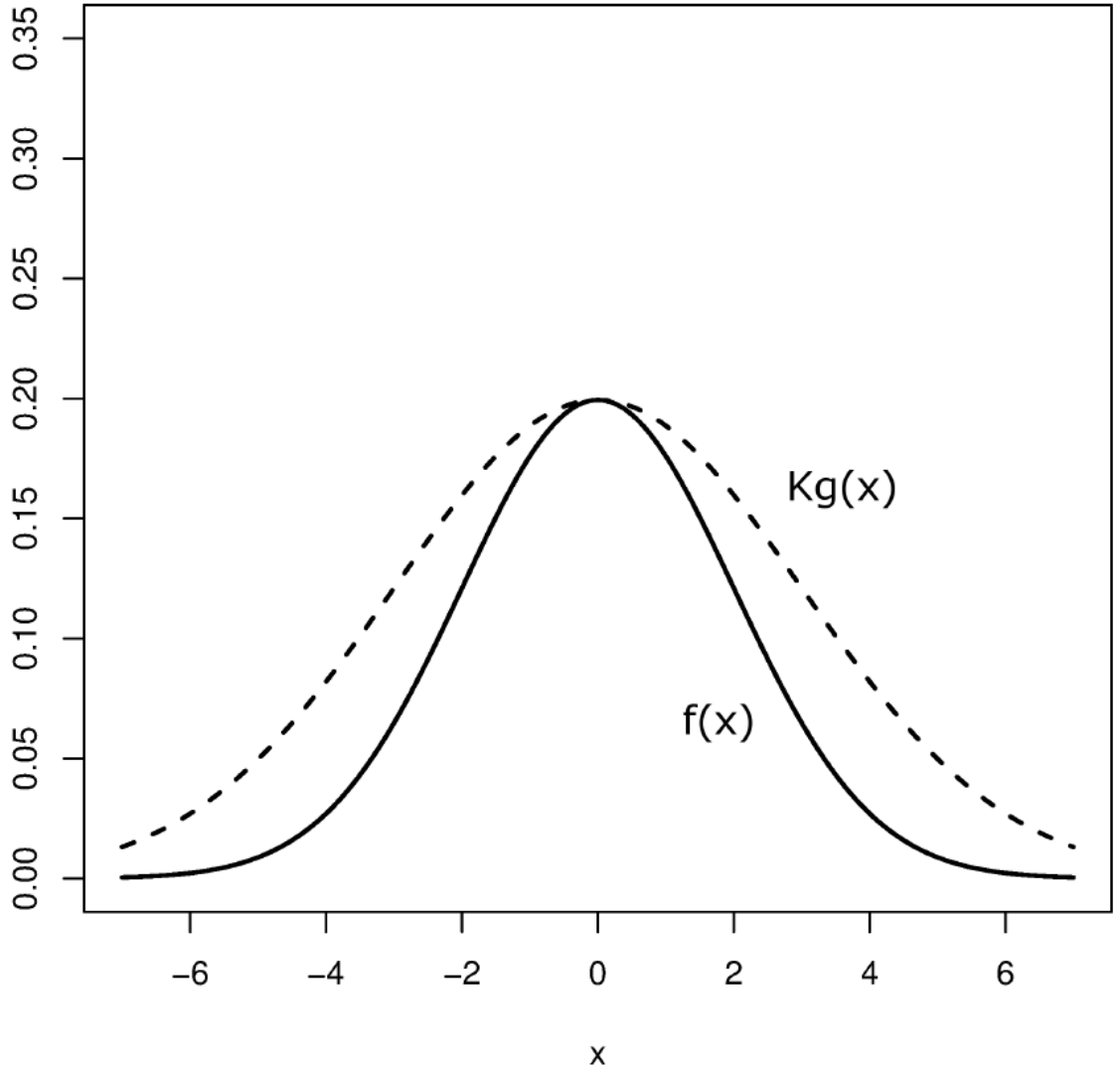
$$P(x) = \int_0^{2\pi} \frac{\ell\,|\cos\theta|}{d}\,\frac{d\,\theta}{2\,\pi}$$
$$= \frac{2\,\ell}{\pi\,d}\int_0^{\pi/2}\cos\theta\,d\,\theta \qquad x \equiv \frac{\ell}{d}$$
$$= \frac{2\,\ell}{\pi\,d}$$

- *π = 2L/Pd*
- https://mste.illinois.edu/activity/buffon/

Normal vs t distributions

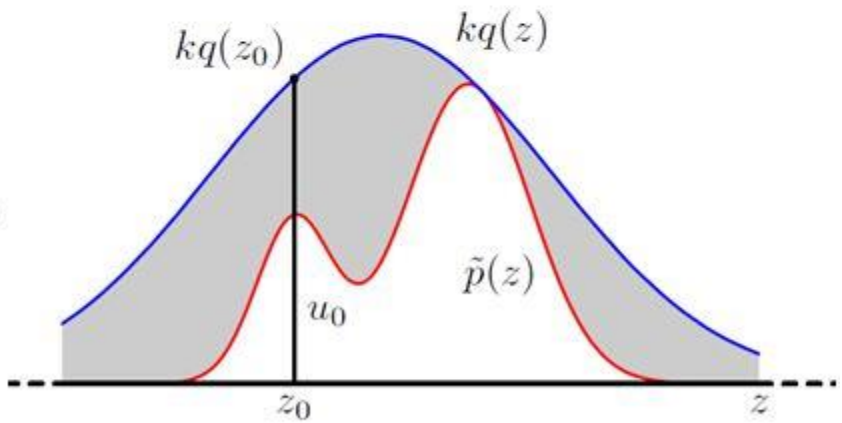


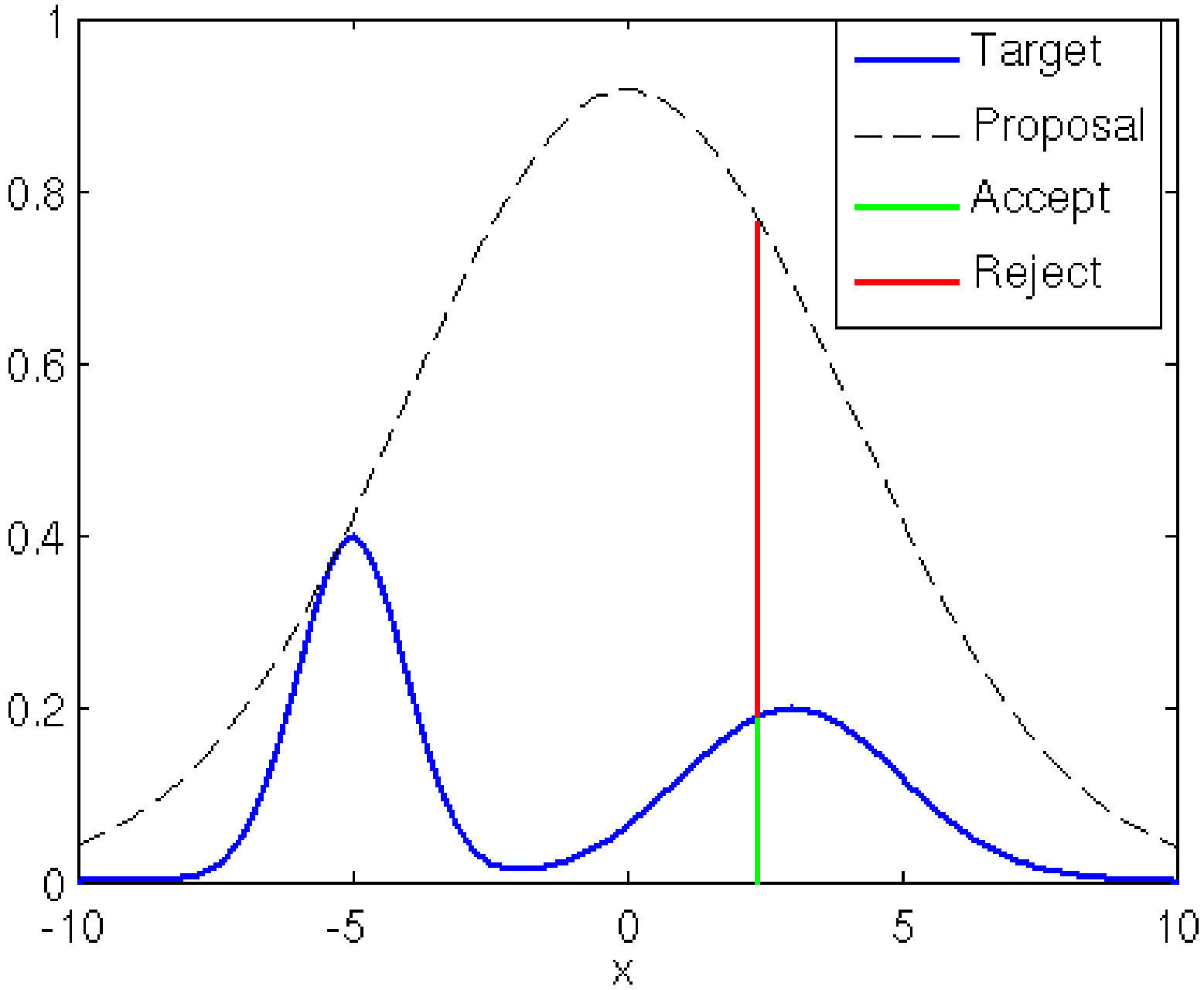
Envelop function

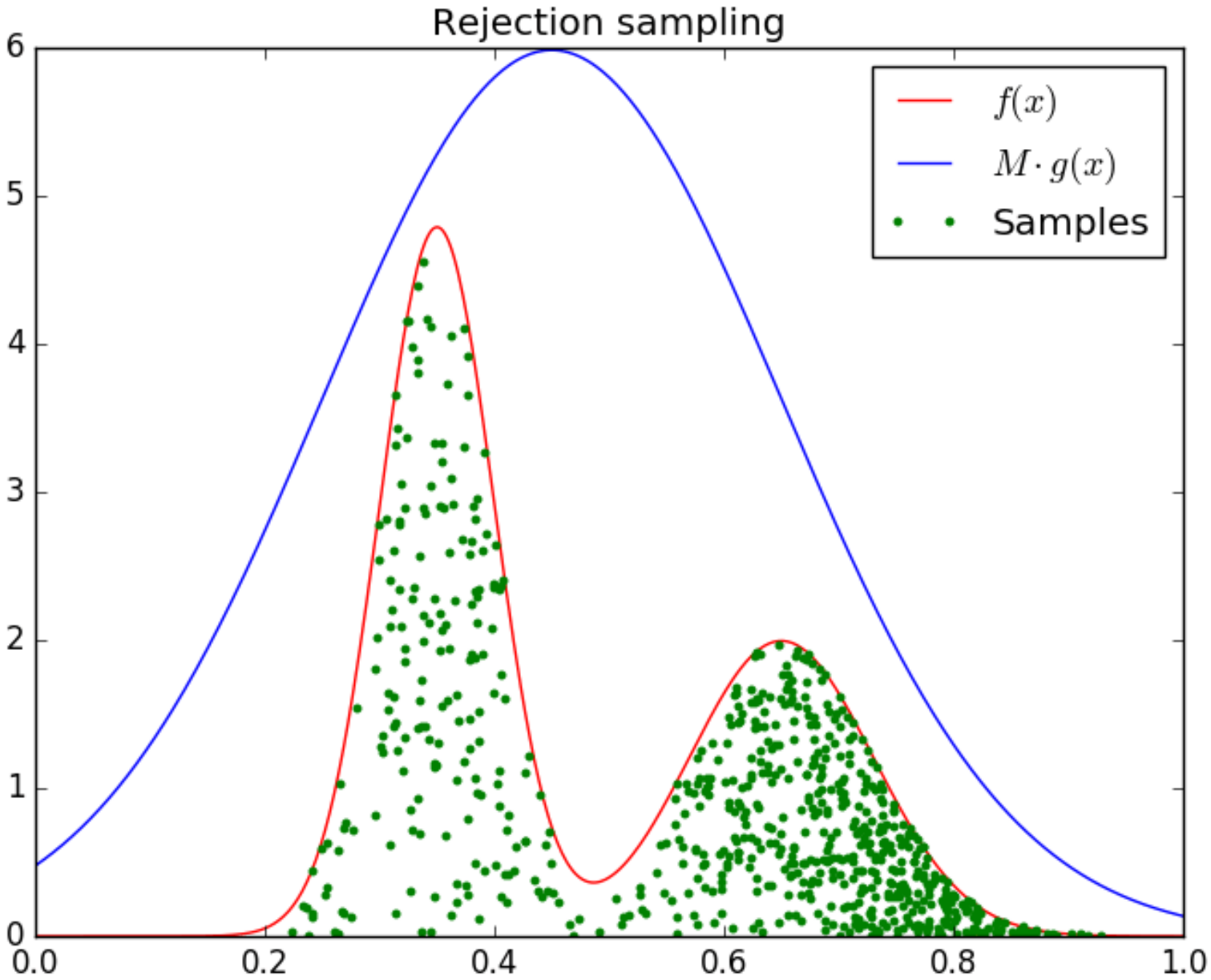


Rejection sampling

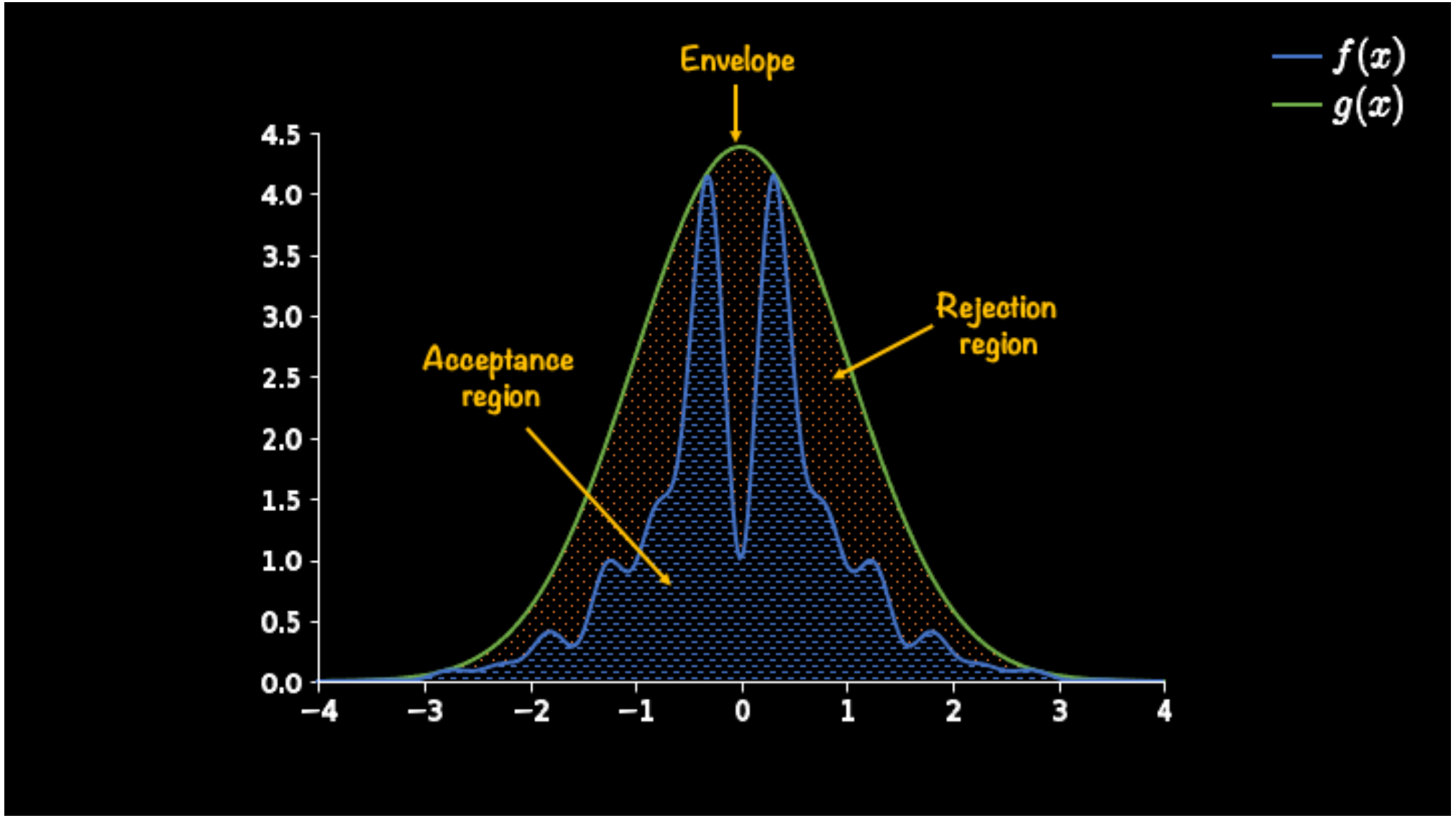
- Rejection sampler
  - Generate two random numbers
  1. number z_0 from proposal distribution $q(z)$
  2. generate a number u_0 from uniform distribution over $[0, \mathrm{k}q(z_0)]$
  - If $u_0 > \tilde{p}(z_0)$ reject!
  - Remaining pairs have unifrom distribution under $\tilde{p}(z)$



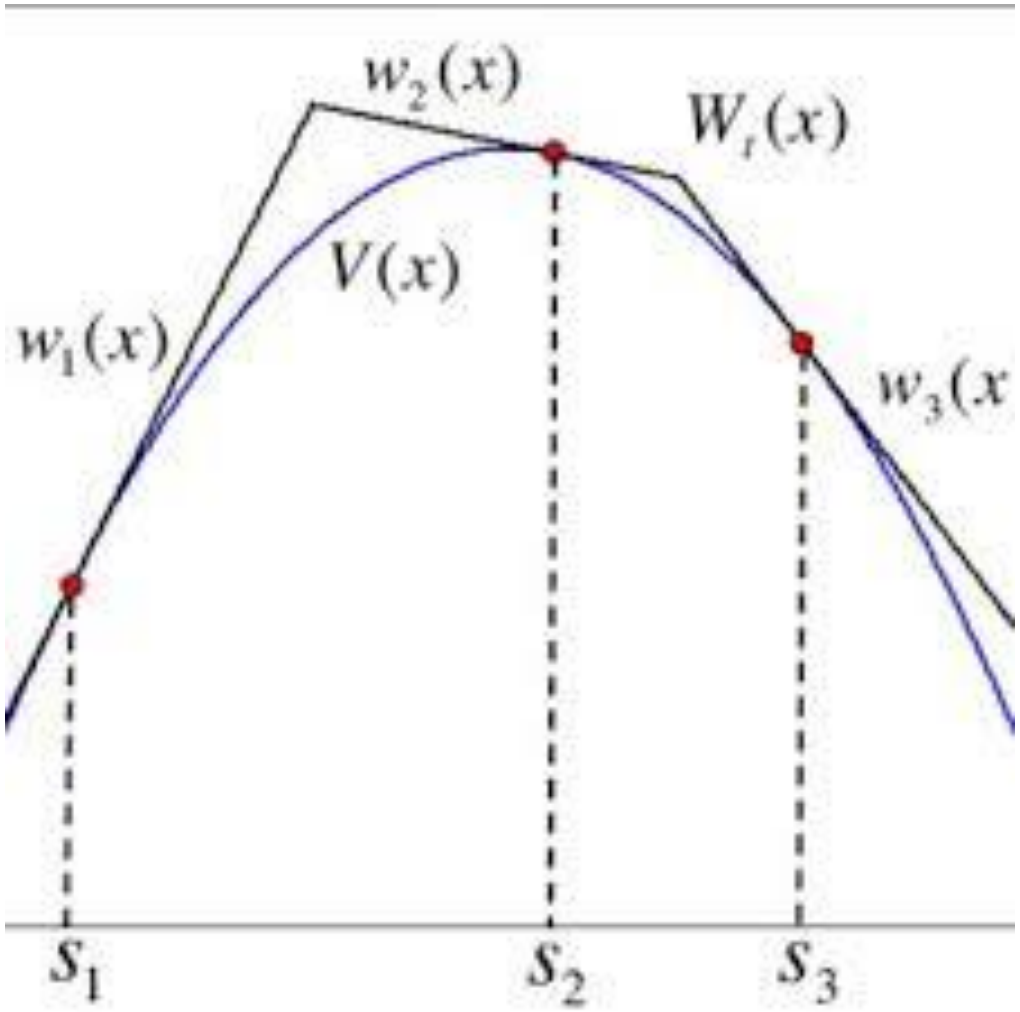

Target
Proposal
Accept
Reject
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10
x


Rejection sampling
$f(x)$
$M \cdot g(x)$
Samples
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Bimodal target function



Adaptive rejection sampling (ARS)



Importance sampling

- Let X be a random variable with PDF p.

$$I = \mathbb{E}(f(X)) = \int f(x)p(x)dx,$$

- We pick a proposal density (also called sampling density) q and generate random numbers $Y_1, \ldots, Y_N$ i.i.d. from q

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^{N} f(Y_i) \cdot \frac{p(Y_i)}{q(Y_i)}.$$

Bias and variance

- It is an unbiased estimator!

$$\begin{aligned}\mathbb{E}(\widehat{I}_N) - I &= \mathbb{E}\left(f(Y_i) \cdot \frac{p(Y_i)}{q(Y_i)}\right) - I \\ &= \int f(y)\frac{p(y)}{q(y)}q(y)dy - I \\ &= \int f(y)p(y)dy - I = 0.\end{aligned}$$

How about the variance?

- Minimize the variance

$$\begin{aligned}
\mathsf{Var}(\widehat{I}_N) &= \frac{1}{N}\mathsf{Var}\left(f(Y_i)\cdot\frac{p(Y_i)}{q(Y_i)}\right) \\
&= \frac{1}{N}\left\{\mathbb{E}\left(f^2(Y_i)\cdot\frac{p^2(Y_i)}{q^2(Y_i)}\right) - \underbrace{\mathbb{E}^2\left(f(Y_i)\cdot\frac{p(Y_i)}{q(Y_i)}\right)}_{I^2}\right\} \\
&= \frac{1}{N}\left(\int\frac{f^2(y)p^2(y)}{q(y)}dy - I^2\right).
\end{aligned}$$

Cauchy-Schwarz inequality

$$\int A^2(y)dy \int B^2(y)dy \geq \left(\int A(y)B(y)dy \right)^2$$

Identifying $A^2(y) = \frac{f^2(y)p^2(y)}{q(y)}$ and $B^2(y) = q(y)$, we have

$$\int \frac{f^2(y)p^2(y)}{q(y)}dy \underbrace{\int q(y)dy}_{=1} \geq \left(\int \frac{f\ (y)p\ (y)}{q(y)^{1/2}} q(y)^{1/2} dy \right)^2 = I^2.$$

Optimal q

$$\mathsf{Var}(\widehat{I}_{N,\mathsf{opt}}) = \frac{1}{N}\left(I^2 - I^2\right) = 0,$$

$$\sqrt{\frac{f^2(y)p^2(y)}{q_{\mathsf{opt}}(y)}} = A(y) \propto B(y) = \sqrt{q_{\mathsf{opt}}(y)},$$

$$q_{\mathsf{opt}}(y) \propto f(y)p(y)$$